

datascope

# Data Quality Diagnostic Report

Source: sample\_sales.xlsx

September 02, 2026

v2.4.0

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**0**

Critical

**2**

Warning

**3**

Info

5 findings across 15 columns and 500 rows

## Executive Summary

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No critical issues were found, but 2 warnings and 3 informational observations were detected. Warnings at this level typically skew joins and aggregations quietly — address them before this data feeds reporting.

## Detailed Findings

### Warning (2)

| Warning         | mostly_null                                                                                                                                                                                                      | Missing Values |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Assumption      | Column 'mostly_null' is expected to be fully populated.                                                                                                                                                          |                |
| Reality         | However, 412 of 500 rows (82.4%) are null or blank. The missing values are scattered across the dataset.                                                                                                         |                |
| Impact          | With more than half its values missing, 'mostly_null' is unreliable for analysis. Aggregations will silently exclude the missing rows, and any model trained on this column will learn from a biased sample.     |                |
| Recommended Fix | Investigate why 'mostly_null' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system. |                |
| Prevention Rule | Add a NOT NULL or completeness check for 'mostly_null' at ingestion time. Flag any batch where null rate exceeds the historical baseline.                                                                        |                |

| Warning         | notes                                                                                                                                                                                                      | Missing Values |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Assumption      | Column 'notes' is expected to be fully populated.                                                                                                                                                          |                |
| Reality         | However, 250 of 500 rows (50.0%) are null or blank. The missing values are scattered across the dataset.                                                                                                   |                |
| Impact          | With more than half its values missing, 'notes' is unreliable for analysis. Aggregations will silently exclude the missing rows, and any model trained on this column will learn from a biased sample.     |                |
| Recommended Fix | Investigate why 'notes' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system. |                |
| Prevention Rule | Add a NOT NULL or completeness check for 'notes' at ingestion time. Flag any batch where null rate exceeds the historical baseline.                                                                        |                |

### Info (3)

| Info            | constant_col                                                                                                                                                                          | Near-Constant Column |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Assumption      | Column 'constant_col' is expected to carry meaningful, varying data.                                                                                                                  |                      |
| Reality         | However, only 1 distinct value was found across 500 rows (uniqueness ratio: 0.2%). Most common: 'FIXED' (500 times).                                                                  |                      |
| Impact          | A near-constant column like 'constant_col' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.           |                      |
| Recommended Fix | Verify whether 'constant_col' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform. |                      |
| Prevention Rule | If 'constant_col' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.                                                  |                      |

| Info            | discount_pct                                                                                                                                                                                                      | Missing Values |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Assumption      | Column 'discount_pct' is expected to be fully populated.                                                                                                                                                          |                |
| Reality         | However, 80 of 500 rows (16.0%) are null or blank. The missing values are scattered across the dataset.                                                                                                           |                |
| Impact          | Missing values in 'discount_pct' will be silently excluded from calculations. If the missingness is not random, aggregations and models will be biased toward the non-missing subset.                             |                |
| Recommended Fix | Investigate why 'discount_pct' has missing values. If the blanks represent a known condition, consider a default value or a separate status column. If they are data-entry gaps, backfill from the source system. |                |
| Prevention Rule | Add a NOT NULL or completeness check for 'discount_pct' at ingestion time. Flag any batch where null rate exceeds the historical baseline.                                                                        |                |

| Info            | mostly_null                                                                                                                                                                          | Near-Constant Column |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Assumption      | Column 'mostly_null' is expected to carry meaningful, varying data.                                                                                                                  |                      |
| Reality         | However, only 1 distinct value was found across 88 rows (uniqueness ratio: 1.1%). Most common: 'value' (88 times).                                                                   |                      |
| Impact          | A near-constant column like 'mostly_null' adds no analytical value. Including it in models or reports may mislead readers into thinking the field varies when it does not.           |                      |
| Recommended Fix | Verify whether 'mostly_null' should actually vary. If not, document it as a constant and consider removing it from analysis. If it should vary, investigate why the data is uniform. |                      |
| Prevention Rule | If 'mostly_null' is supposed to carry diverse values, add a data-quality check that flags columns with fewer than 1% unique values.                                                  |                      |

# Field Inventory

Summary of all fields with detected issues.

| Field Name   | Issue Types                          | Highest Severity | Finding Count |
|--------------|--------------------------------------|------------------|---------------|
| constant_col | Near-Constant Column                 | Info             | 1             |
| discount_pct | Missing Values                       | Info             | 1             |
| mostly_null  | Missing Values, Near-Constant Column | Warning          | 2             |
| notes        | Missing Values                       | Warning          | 1             |